Stylized Racing Track Set

Thank you for purchasing the **Stylized Racing Track Set**! This document will guide you through how to use the assets, along with some tips to get the most out of them.

1. What's Included

This asset pack contains a wide collection of stylized, modular racing track elements designed for quickly building race circuits and environment scenes. All props share a consistent stylized visual design and are optimised for easy scene building.

The pack contains the following:

Modular Racing Tracks

A large set of modular road pieces that can be combined to create custom racing tracks and driving environments.

- Straight road segments in multiple heights.
- Curved road segments with different shapes and radii.
- Sloped and elevated track pieces, tunnels.
- Modular and standalone tunnels.
- Start and finish track sections.

All road pieces are designed to connect seamlessly using Unity's grid snapping system, allowing user to quickly build tracks without visible seams.

The road pieces are built with additional thickness rather than being flat planes, allowing them to be placed on slightly uneven terrain without visual clipping.

Track Variations & Borders

Road modules can be used as plain tracks or enhanced with additional decorative and functional elements.

- Track borders with stylized curb markings: white or red/white borders.
- Guard rails and safety fences.
- Elevated track variants with pillars or full concrete support.
- Modular elements for pit stop.

These variations allow you to create anything from simple racing layouts to fully detailed circuits.

Racing Props

Additional props are included to help create lively and complete racing environments.

- Start and finish gates.
- Spectator stands and chairs.
- Pit stop tents and buildings.
- Decorative flags, banners, sound and lighting props.
- Tire barriers, cones, and safety props.
- Seating and small environment decorations

The prefabs are prepared in 4 different team colours which can easily be changed by using different materials.

Nature elements

Stylized nature assets are included to help populate and enhance outdoor racing environments.

- Trees in multiple variations.
- Different rocks.

Trees and rocks are optimised to be used with Terrain.

Example Scenes

All scenes shown on the product page are included in the assets pack:

- Three scenes showcasing how these assets can be combined and used.
- Sample scene with all available prefabs, providing an organised overview of the assets, for easy browsing and preview.

2. Features

Stylized Racing Look

All models are designed in a clean, low-poly stylized aesthetic. Bright colours, smooth shapes, and simplified forms make this pack ideal for arcade racers, casual games, stylized simulators, or playful environments.

The cohesive visual style ensures that all track pieces, props, and environment elements blend seamlessly together.

Modular System

The racing track is fully modular and designed for fast level building.

- Straight, curved, sloped, and elevated road segments.
- Interchangeable borders and guard rails.
- Modular barriers and fences.
- Seamless snapping using Unity's grid system.

All road pieces align cleanly without visible seams when using proper grid snapping.

Additionally, the road modules are modelled with depth instead of being flat planes, making them easier to place on slightly uneven terrain while maintaining clean visuals.

Colliders

All prefabs include custom mesh colliders.

All prefabs include appropriate colliders:
- Road pieces and props use custom mesh colliders.
- Nature elements (trees and rocks) use simple colliders to work with Terrain.

This allows immediate use in gameplay without requiring additional setup.

Textures and Materials

The pack primarily uses solid-colour materials to maintain flexibility and performance.

Material Types:
- Standard materials.
- Alpha transparency materials (for fence and trees).
- Standard texture materials (asphalt, tires, banners and start/finish arches), texture resolution: 1024 × 1024 px and 2048 × 2048 px.
- Optimised for low-poly stylized look.

Most assets use simple materials without heavy texture usage, keeping the pack lightweight and easy to modify.

Number of Materials:

- 4 texture materials for asphalt,
- 10 standard materials for basic elements (road markings, asphalt, concrete, metal)
- 18 standard and 4 texture materials for props and equipment,
- 7 texture materials for environment (grass, sand),
- 2 standard and 8 texture materials for greenery,
- 1 skybox material.

Easily Customisable Colours

Since most assets use non-textured materials, colours can be quickly adjusted directly in Unity.

This makes it easy to:
- create custom team colours,
- adjust track themes,
- match different game environments,
- create day/night or seasonal variations.

Prefab Count

Large collection of modular pieces and variations includes over 250 prefabs, including modular roads, borders, props, and environment elements.

Rendering Pipeline Compatibility:

Built-in Render Pipeline: Yes (native support).

Universal Render Pipeline (URP): materials can be converted using Unity's material upgrade tools (instructions included in documentation).

High Definition Render Pipeline (HDRP): materials can be converted using Unity's material upgrade tools (instructions included in documentation).

3. Installation

1. Open your Unity project.
2. Import the asset package:
   Assets → Import Package → Custom Package.
3. Select the downloaded **Stylized Racing Track Set** .*unitypackage* file.
4. In the import window, make sure all files are selected and click **Import**.
5. After the import is complete, the assets will appear in your **Project** window, including:
   - prefabs,
   - materials and textures,
   - example scenes.

You can now drag and drop the prefabs directly into your scene to start building racing tracks.

4. Folder Structure

Once imported, you'll find the following folders under
Assets > Stylized-Racing-Track-Set:

- **Prefabs:** Contains ready-to-use prefabs for modular roads, borders, racing props, buildings, nature elements, and other environment assets. Prefabs can be dragged directly into your scene to quickly build racing tracks and environments.

- **Materials:** Includes all materials used in the asset pack, such as road materials, standard and texture materials for props and environment, with colour variations.

- **Scenes:** Contains example scenes demonstrating how the assets can be used, including 3 demo racing track scenes and a scene with all prefabs displayed for easy preview.

5. How to Use

Racing Track Pieces

Drag and drop the desired road prefabs directly into your scene.

You can combine different types of track modules to build your racing circuit:
- straight segments with different height,
- curved road pieces with different shapes and sizes,
- sloped and elevated road segments,
- tunnels,
- different borders,
- guard rails.

Tracks can be used as plain roads or enhanced by adding borders, guard rails, or fences.

Borders and Fences

Borders and safety elements can be added to improve both the visual appearance and functionality of the racing track.

User can attach 2 different types of borders along the edges of the road. There are 3 types of prefabs:
- Plain road,
- Standalone borders (2 types),
- Roads with both types of borders as pre-made prefab.

User can add guard rails on elevated sections, use barriers to guide players or block areas, … These elements are modular and designed to align easily with the road system.

There are also end pieces available to add nicer finish the guard rails or the white-red borders when used only on corners.

Racing Props

Drag and place racing props around the track to create a more dynamic environment.

Pack includes:
- start and finish structures,
- flags and banners,
- cones and tire barriers,
- lighting and sound elements,
- pit stop elements,
- tribunes.

These props help bring life and detail to the racing scene.

Nature Elements

Trees, rocks, and vegetation can be placed around the track to build a complete outdoor environment.

For example:
- line the track with trees,
- add rocks for terrain variation,
- fill empty spaces with vegetation.

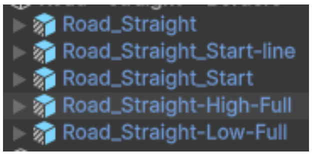
Trees and rocks are optimised to be used with Unity's Terrain element.

Naming

Prefabs are organized to make building tracks fast and intuitive, even for beginners. Prefab names include descriptors for size, type, or variation, making it easier to find the correct asset.

Modular road and track elements follow a consistent naming structure to help with quick assembly and organization.

Example:



6. Example Scenes

To help you get started with the Stylized Racing Track Set, several example scenes are included in the Scenes folder. These scenes demonstrate how the modular track system, props, and environment elements can be combined to create complete racing environments.

- **Example Track Scenes**
  These scenes showcase different racing track layouts built using the modular road system. They demonstrate how track pieces, borders, props, and nature elements can be combined in practical setups. Scenes are designed to help you quickly understand the modular workflow and start building your own racing tracks.

- **Prefab Overview Scene**
  This scene displays all prefabs organized by category for easy browsing and reference.
  It allows you to quickly preview available assets and test different pieces before placing them into your own scenes.

7. Tips for Customisation

Materials and Colours

Most assets in this pack use simple materials without heavy textures, making it easy to customize their appearance.
You can:

- Duplicate a material and change its colour.
- Create different track themes (e.g. night track, desert track, neon track).
- Adjust borders, props, or environment colours to match your game style.

This allows you to quickly adapt the assets to different environments or art directions.

Textures

You can modify the look of banners by duplicating the material and changing its albedo texture.

Track Layout Variations

The modular system allows you to create a wide variety of racing tracks and to:

- Combine straight and curved segments to design unique layouts.
- Add slopes, tunnels and elevated sections for more dynamic tracks.
- Use borders, fences, and barriers to shape the course.

Experimenting with different combinations helps create diverse racing environments.

Environment Customisation

Props and nature elements can be used to give each track its own identity.

For example:

- add trees and rocks around the track,
- place banners, lights, and racing props along the course,
- create a pit stop with custom number of pit boxes, with tents or garage buildings,
- add rocks, green or autumn trees, or palms,
- create spectator or decorative areas near the track.

This helps transform a simple track into a complete racing scene.

8. Troubleshooting

Missing Materials:

If any prefab appears to be missing materials or textures:

1. Ensure that the **Materials** folder is imported correctly.
2. Try **Reimporting** the asset package if needed.

The Assets pack was made with **Built-in Render Pipeline**.
In case you are using Universal Render Pipeline (URP) or High Definition Render Pipeline (HDRP), materials need to be converted. Instructions for converting materials can be found in chapter "**Material conversion for render pipeline compatibility**".

Prefabs Not Snapping Correctly

If track pieces are not aligning properly:

1. Check your **Snap Settings**:
   Go to "Edit → Grid and Snap Settings" and adjust the grid size to match the prefab dimensions.
2. Make sure the Move Tool is set to:
   - Pivot
   - Global
3. Enable **Grid Snapping** when moving objects (hold Ctrl on Windows or Cmd on Mac while moving).

Road Pieces Not Aligning Perfectly

Ensure that:

- Rotation is set to 0 / 90 / 180 / 270 degrees.
- The objects are aligned to the grid before connecting additional pieces.

4. Material Conversion - Render Pipeline Compatibility

Before starting, ensure you've backed up your project. Switching render pipelines will modify materials and potentially other settings, so it's recommended to keep a copy of your original project.

I. Converting to the Universal Render Pipeline (URP)

Unity provides an automated material converter for URP.

Method 1 – Using the Render Pipeline Converter (recommended in newer Unity versions)**:**

1. Install Universal Render Pipeline from the Package Manager if it is not already installed.
2. Go to:
   Window → Rendering → Render Pipeline Converter.
3. Select Built-in to URP conversion.
4. Run the converter to upgrade materials.

After the conversion process finishes, review your materials and scenes to ensure everything appears correctly. You may need to adjust some material properties such as Smoothness.

Method 2 – Using the legacy menu option:

Go to:
Edit → Render Pipeline → Universal Render Pipeline → Upgrade Project Materials to UniversalRP Materials.

This will convert all materials in your project to URP-compatible versions.

Alternatively, to convert only selected materials, select them in the Project Window and choose Upgrade Selected Materials instead.

Check the materials to ensure they look correct. URP uses a slightly different material model than the Built-in Pipeline, so you may need to adjust Smoothness or other properties for the best results.

II. Converting to the High Definition Render Pipeline (HDRP)

Unity also provides an automated material converter for HDRP.

Method 1 – Using the Render Pipeline Converter (recommended in newer Unity versions)**:**

1. Install High Definition Render Pipeline from the Package Manager if it is not already installed.
2. Go to:
   Window → Rendering → Render Pipeline Converter.
3. Select Built-in to HDRP conversion.
4. Run the converter to upgrade materials.

HDRP materials support additional advanced features such as Detail Maps, Translucency, and Clear Coat.

Method 2 – Using the legacy menu option:

Go to:
Edit → Render Pipeline → High Definition Render Pipeline → Upgrade Project Materials to High Definition Materials.

This will convert all materials in the project to HDRP-compatible versions.

Alternatively, to convert only selected materials, select them in the Project Window and choose Upgrade Selected Materials.

HDRP materials offer additional features, such as Detail Maps, Translucency, and Clear Coat options, which aren't available in the Built-in Pipeline.

Important Note

After conversion, review your prefabs and scenes to ensure materials were applied correctly.
In some cases, you may need to manually reassign materials if the automatic conversion missed them.